# COMMUNICATION BETWEEN PSYCHOPY AND UNICORN: LSL TRIGGERS

## Psychopy setup

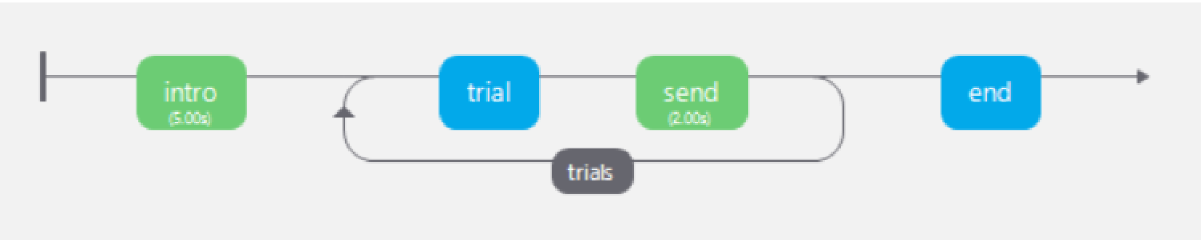
https://mbraintrain.com/how-to-send-lsl-triggers-with-psychopy-and-mobile-eeg/

1. Install the pylsl in order to make this LSL code work.

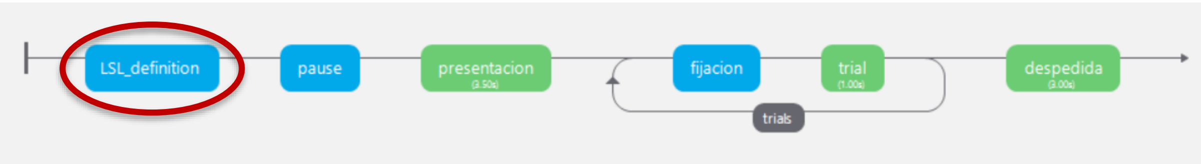
    pip install pylsl

    conda install -c conda-forge liblsl

2. Prepare a computerized cognitive experiment with PsychoPy (p.e. Motor Imagery):

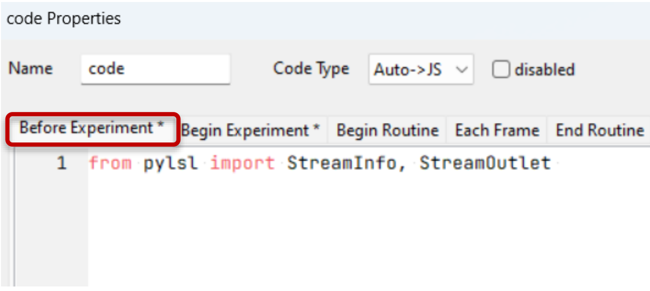


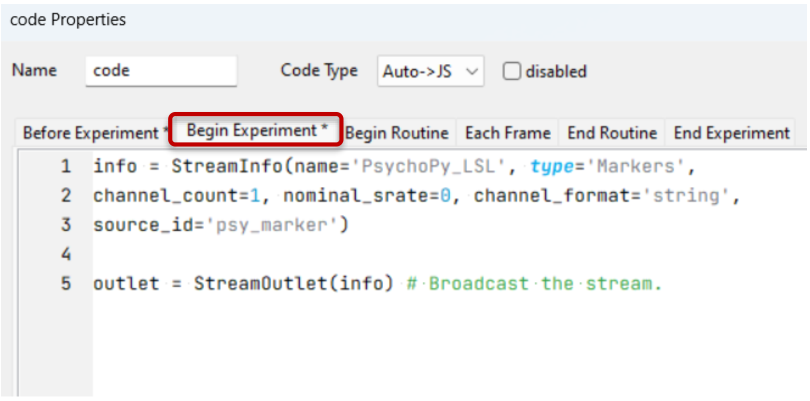

3. Insert a new routine where we can define the LSL stream:



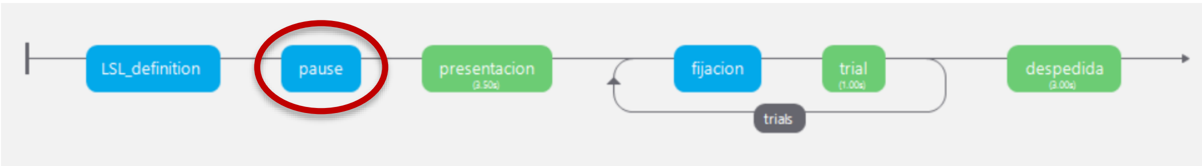

4. In this routine, add block of code.


5. Double click on “code” and define the LSL stream by indicating the next phyton code on the sections “Before experiment” and “Begin experiment”:



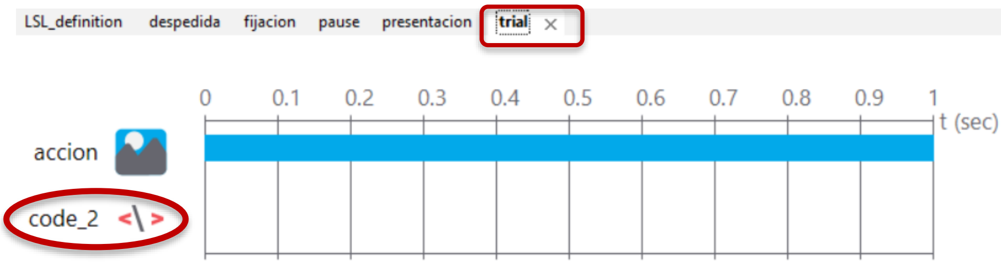



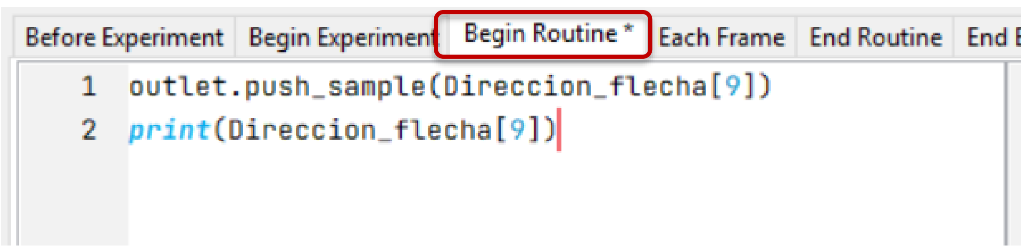

6. Introduce a pause between LSL definition and the rest of the computarized task:




7. Add another block of code whenever the stimulus is sent and write the following code under the “Begin Routine” tab (SEND).

*Outlet.push_sample(['event_name'])*

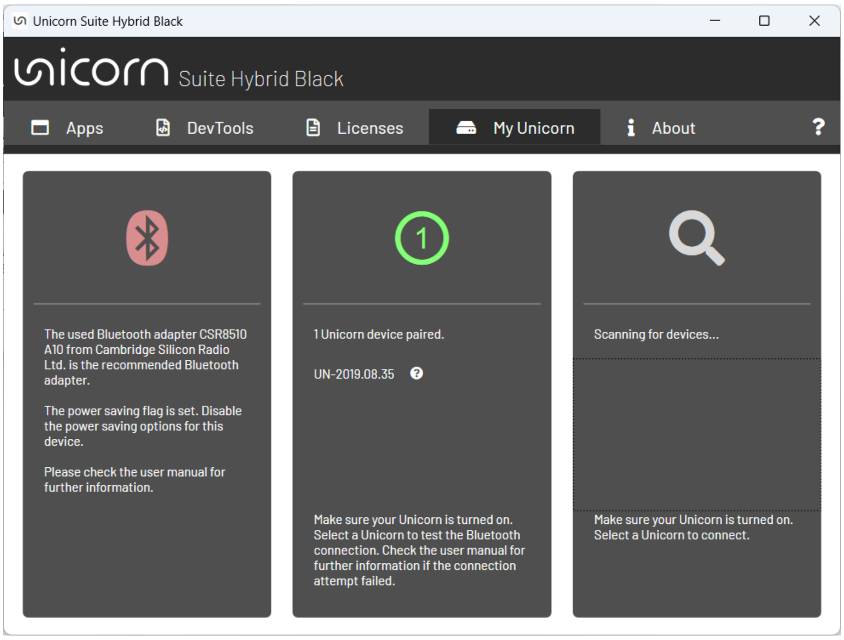


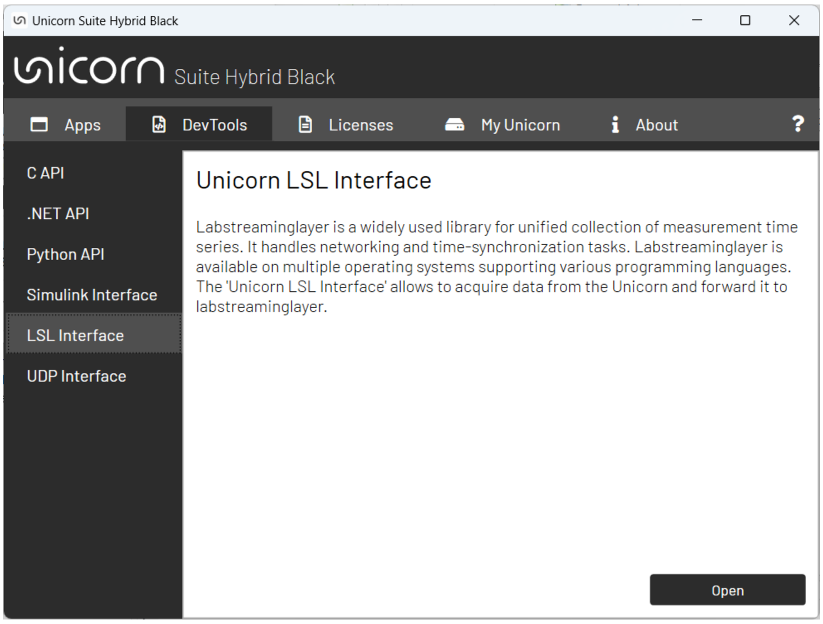
8. Make sure there is a pause after defining the LSL and before the experiment begins so that LabRecorder can detect the stream correctly. When LabRecorder has begun, then continue.

### Unicorn LSL Setup

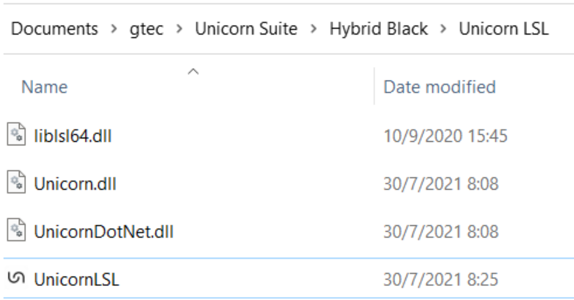
1. Open Unicorn Suite software and pair the Unicorn device to the laptop via Bluetooth (use adapter).



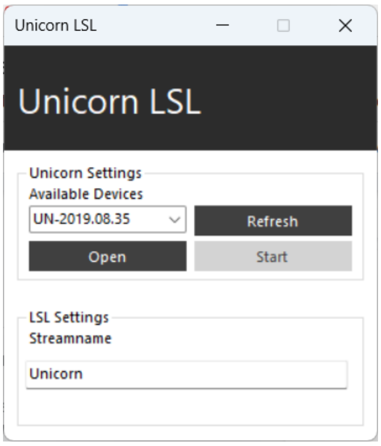

2. Under *DevTools*, select *LSL Interface* and press *Open*.



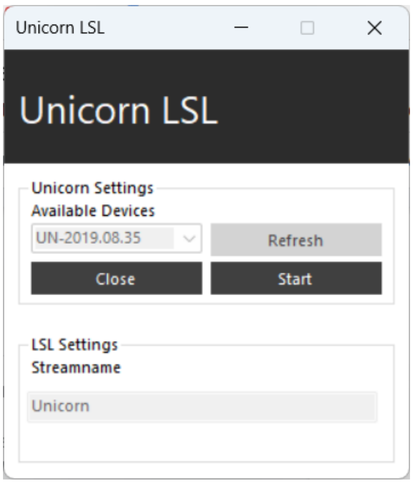

3. Execute *UnicornLSL* file.




4. Select the Unicorn device and set a stream name for the EEG signals. Then, press *Open*.

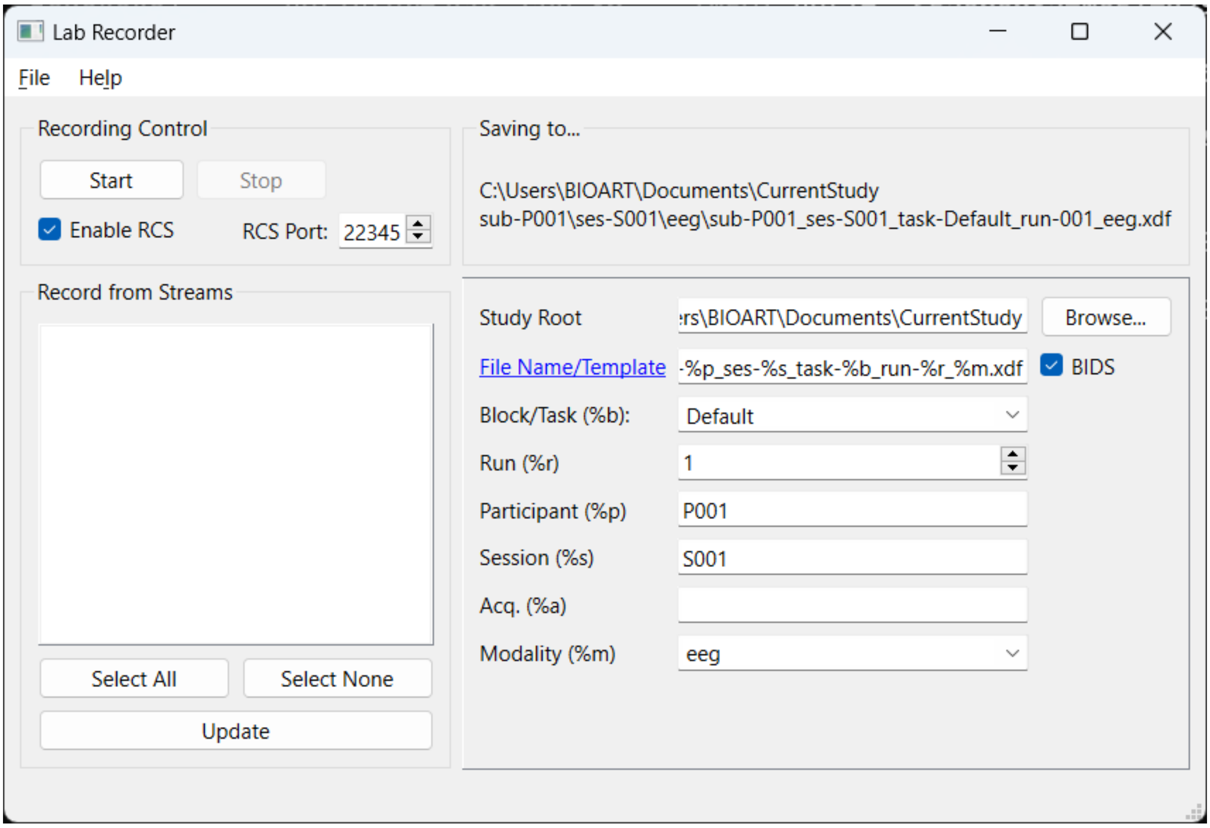



5. Start streaming.




### Lab Recorder

1. Install software https://github.com/labstreaminglayer/App-LabRecorder/releases
2. A window like this will show up




3. Select streams to record (both events and eeg streams MUST be opened) and press *Start*.
4. When pressing *Stop*, an .xdf file will be saved on the *StudyRoot* path.

### Matlab Processing

1. Use this module to open the dataset with Matlab [https://github.com/xdf-modules/xdf-Matlab/tree/0cdf054391fff7f0ea3416ee632ad1bd73d6623b]
    - I have upload the required script ('load_xdf.m') on Atenea.

2. *[streams,fileheader] = load_xdf('name_eeg_data.xdf')*